MPSTME – NMIMS
MBA TECH (Data Science)

Microeconomic Analysis of the Indian Iron & Steel Industry

A Firm Level Study of Tata Steel

Aditya Mishra, S046
Yashas Madan, S040

Abstract

The research investigates the Indian iron and steel industry through microeconomic analysis by examining Tata Steel at the company level. The industry operates as a concentrated oligopoly because the combination of high capital requirements and limited raw material access and the advantages of economies of scale creates substantial barriers that prevent new companies from entering the market while established companies maintain strategic relationships with their main competitors. The Indian steel market derives its demand primarily from the infrastructure and construction and automotive industries which show price stability in the short term but display strong income responsiveness that correlates with both GDP growth and urbanization. The company incurs most of its operational expenses from variable input costs which include iron ore and coking coal, and this leads to decreased profitability when global commodity prices change, while the company needs to maximize its production output to meet its fixed costs requirements. The government implements capacity expansion programs and competitiveness initiatives through its National Steel Policy 2017 and Production Linked Incentive Scheme programs. The research demonstrates that companies can achieve sustained growth and establish themselves as market leaders through their cost management practices and innovative methods and their environmentally friendly transition approaches which they implement in response to changing global trade policies.

Overview of the Indian Iron & Steel Industry

Recent archaeological investigations conducted in the Middle Ganges Valley demonstrate that iron working in India started to develop during the period of 1800 BCE (1800 years before the birth of Christ). The first metal manufacturing operation in India began to produce useful metal products. The industry has been essential for regional growth since the beginning of tool making activities. This industry serves as the fundamental engine that drives our economic progress through its development of infrastructure and defense systems and industrial operations. The Indian Iron & Steel Industry evolved from TISCO (Tata Iron & Steel Company) which started operations in 1907. The economy currently relies on this organization as one of its key economic pillars. The industry operates integrated steel plants together with secondary producers and ancillary units which create value for GDP and exports and employment. The industry started its journey from TISCO but experienced multiple unsuccessful attempts before reaching this point which included the failed Port Novo operation in 1830. J. Nehru established Hindustan Steel Limited (HSL) in 1954, after independence which enabled public-sector plants to build facilities in Bhilai and Durgapur and Rourkela during the Second Five-Year Plan which later unified under Steel Authority of India Limited (SAIL) in 1973. India became the world's second largest steel producer after Japan in 2018. The industry maintained its growth during the time period from 2019 to 2023 with a CAGR of 6%◦.China showed minimal growth of 1% during this period, while the global market experienced a decline of 1%. Total finished steel consumption increased from 100.171 MT in 2019-20 to 136.291 MT in 2023-24, showing strong domestic demand which achieved growth of 13.7% compared to last year. The steel sector in India will fulfill all future demand requirements which will happen during the upcoming fiscal years according to market predictions that include a conservative 6% CAGR through fiscal year 2027. The current iron and steel industry of India exhibits a lively atmosphere which presents challenges because of its labor laws and environmental regulations. The industry between JSW and Tata encounters major conflicts between its dominant players because of their aggressive lobbying activities and competitive bidding procedures. The industry becomes difficult to manage because of high inventory days and cyclical work patterns and extensive CapEx investments. The iron and steel industry faces operational difficulties because of its inventory management and its need for major capital expenditures which occur in cyclical work periods. Public-sector SAIL operates flagship plants in Bhilai and Bokaro, while Tata Steel and JSW Steel and JSW Steel dominate the market through their advanced facilities. Emerging leadership positions belong to ArcelorMittal Nippon Steel and AM/NS India, which focus on producing green steel while secondary producers like Essar Steel and Jindal Steel & Power manufacture long products. The top five companies control more than 50% of production with FDI attracting international companies because of the 100% foreign investment policy.Demand from infrastructure development (roads, railways) and real estate and automobile and appliance industries which will reach 300 million tonnes by 2030 will grow because of urbanization and PM Gati Shakti initiatives. Supply has expanded quickly with crude steel production reaching ~140 million tonnes in FY2024 but the sector faces import challenges during the monsoon season and when raw materials become scarce.

Market Structure Analysis

The Indian iron and steel market operates as an oligopoly because it has a few major integrated producers who control the entire market. Price leadership in the market gets demonstrated through the business activities of JSW Steel and Tata Steel because their pricing and capacity expansion and raw material sourcing decisions create strategic reactions from their competitors, which proves that they operate in shared dependence typical to an oligopoly. The Steel Authority of India Limited and Jindal Steel and Power and ArcelorMittal Nippon Steel India together with their respective output production power, create a market situation, which results in a high Herfindahl-Hirschman Index measurement, which shows a dominant market position, which we use to determine the presence of limited market competition. In this business environment, enterprises show restricted price competition behavior while their operations adopt non-price competition methods through technological advancements, plant digitalization, environmental sustainability programs, and product development activities that include advanced high-strength steel products and coated sheet materials and specialized electrical steel products that serve automotive and infrastructure uses. The process of building integrated blast furnace or DRI-EAF facilities requires massive financial investment that reaches billions of dollars, coupled with extended operational downtime, environmental and land acquisition approval requirements, and essential access to proprietary iron ore and coking coal resources, which creates substantial obstacles for newcomers while sustaining existing advantages that come from scale benefits and shipping operations and brand partnerships. Firms have developed an oligopoly market structure through their strategic activities which include backward integration into mining operations to control input costs and joint technology development partnerships and export hedging activities as well as their sustainability efforts to meet emerging policy requirements and investor sustainability expectations through green hydrogen testing projects and carbon footprint reduction strategies. The government uses policy tools such as import duties to create indirect protection for domestic producers, which helps sustain the market position of major companies. India’s iron and steel industry maintains an existing oligopoly market structure because its enterprises establish interdependencies through their business practices, which create high industry concentration, and they face major obstacles to entering the market, and they utilize both direct and indirect strategic communication methods to maintain their market power.

Demand Analysis

India obtains its primary steel requirements through derived demand because the material is essential for construction work which requires steel as an intermediary material in key sectors that include infrastructure projects and building construction and automobile production and engineering activities. The total demand for infrastructure projects reaches 40 to 45 percent according to our findings which identify that public investments through PM Gati Shakti programs create the primary driving force for demand while real estate and housing generate 25 percent and automotive vehicles produce 15 percent of demand and the remaining demand originates from machinery and consumer products and capital equipment.

The projected demand growth for 2026 will approach 8 to 9 percent because urbanization and continuing government capital expenditures will drive demand growth for these sectors which include housing construction and automotive manufacturing and machinery production. The analysis of price elasticity of demand reveals short-term inelastic demand which exists between -0.2 and -0.5 because buyers who work in infrastructure development and original equipment manufacturing choose to base their purchasing decisions on supply reliability and execution speed instead of the slight price changes which occur in the market. The long-term elasticity of the system shows a slight increase because people can select between different options while special applications in automotive production permit aluminum to substitute steel. The use of aluminum in certain automotive applications enables material replacement yet infrastructure together with construction relies on steel for its essential structural components.

The income elasticity of demand demonstrates a strong positive relationship which exceeds one at approximately 1.5 to 2.0 because our demand for housing and vehicles and appliances increases when GDP growth exceeds 7% and household incomes rise which causes our steel consumption to increase at an accelerated pace. Cross elasticity maintains a positive relationship with substitutes such as aluminum or plastics yet displays a lower value of approximately 0.3 which shows gradual sectoral shifts instead of sudden material transitions. Cement works function as complements because cement expansion in construction projects creates a need for both essential construction materials and cement.

Elasticity analysis shows that India maintains its steel demand pattern, which exists at a constant level, throughout the short term because its derived demand structure and project-based requirements. The system responds to income increases while showing only limited effects from people switching between different product options. Demand will stabilize because industrial activities and infrastructure projects continue to expand.

Cost Structure and Production Economics

The steel industry in India possesses a cost structure which derives its primary components from raw materials and energy consumption yet fixed costs serve as the main determinant for the company's profit margins and operational efficiency and production capacity. The company spends approximately 40 to 50 percent of its total budget on fixed costs which result from the depreciation expenses for blast furnaces and coke ovens and sinter plants and land and rolling mills and the considerable debt service requirements for capacity expansion projects. Total costs divide into variable costs which make up 50 to 60 percent of expenses while raw materials account for the largest percentage because iron ore comprises 25 to 30 percent and coking coal makes up 35 to 40 percent and energy and power costs reach 15 to 20 percent and labor and logistics expenses occupy 10 to 15 percent while logistics costs typically take up 16 to 18 percent of revenue because of India's freight system and extended supply chains.

The analysis of economies of scale shows that integrated plants which have production capacities between 5 and 10 million tonnes achieve 20 to 30 percent lower costs per tonne compared to mini-mills which operate at smaller capacities because they obtain raw materials through bulk purchasing and use their own mining facilities and implement waste heat recovery systems and manage their energy consumption effectively. The industry uses capacity utilization as its main factor to determine profit margins because the sector will operate at 75 to 80 percent capacity in 2026 which falls short of the 85 to 90 percent capacity range that produces maximum operational efficiency.

The domestic capacity expansions together with an 8 percent demand increase establish a situation where oversupply occurs because cheaper imports from China and high-cost plants force smaller production facilities to stop operations. The sector experiences extreme vulnerability to fluctuations in input prices because a 10 percent increase in coking coal prices results in additional steel production costs of approximately ₹2,000 to 3,000 per tonne and NMDC Limited's iron ore price increases of ₹150 to 250 per tonne directly impact profit margins because producers pass through 70 to 80 percent of cost increases to buyers in markets where they can do so.

The industry still faces structural risks from global commodity cycles despite energy reforms and renewable energy sources and higher scrap recycling rates in electric arc furnaces. The analysis of cost components in production shows that the Indian steel sector needs to focus on scale efficiency and input cost management and high utilization rates to establish competitive advantages and achieve sustained performance.

Government Intervention and Regulatory Framework

The government of India controls the steel industry through its regulatory framework which includes protective trade restrictions and environmental regulations and industrial development programs that maintain local industry protection while enabling international competitiveness. The government has established safeguard duties which range from 11 to 12 percent for certain flat steel imports to control import surges from China Vietnam and South Korea until the duties will be reduced in 2026.

The current anti-dumping investigations focus on particular product categories which aim to protect against predatory pricing practices while the export strategies work to establish preferential quota arrangements with the European Union through the various overcapacity frameworks and prepare domestic producers for the Carbon Border Adjustment Mechanism (CBAM) which will introduce extra carbon costs between January 2026 and ₹2,000–3,000 per tonne on all emission-intensive goods.

The Green Steel Mission and its environmental regulations enforce stricter rules which mandate companies to achieve 10 to 15 percent emission reductions by 2030. The industrial development policies create pathways for growth through their Production Linked Incentive Scheme which dedicates ₹6 000 crore to developing specialty steel and their National Steel Policy 2017 which enables 100 percent FDI to reach a total capacity of 300 million tonnes by 2030. The infrastructure projects which the government implements under PM Gati Shakti boost demand for construction work while the logistics system improvements which aim to decrease freight charges between 10 to 14 percent will enhance the ability of businesses to compete.

The assessment of these interconnected policy measures which include trade protection to prevent unfair imports and environmental regulation for sustainable development and industrial incentives for capacity building demonstrates that the government designed a strategic system to safeguard domestic industries while expanding India's position in global steel manufacturing networks.

Firm-Level Microeconomic Analysis: Tata Steel

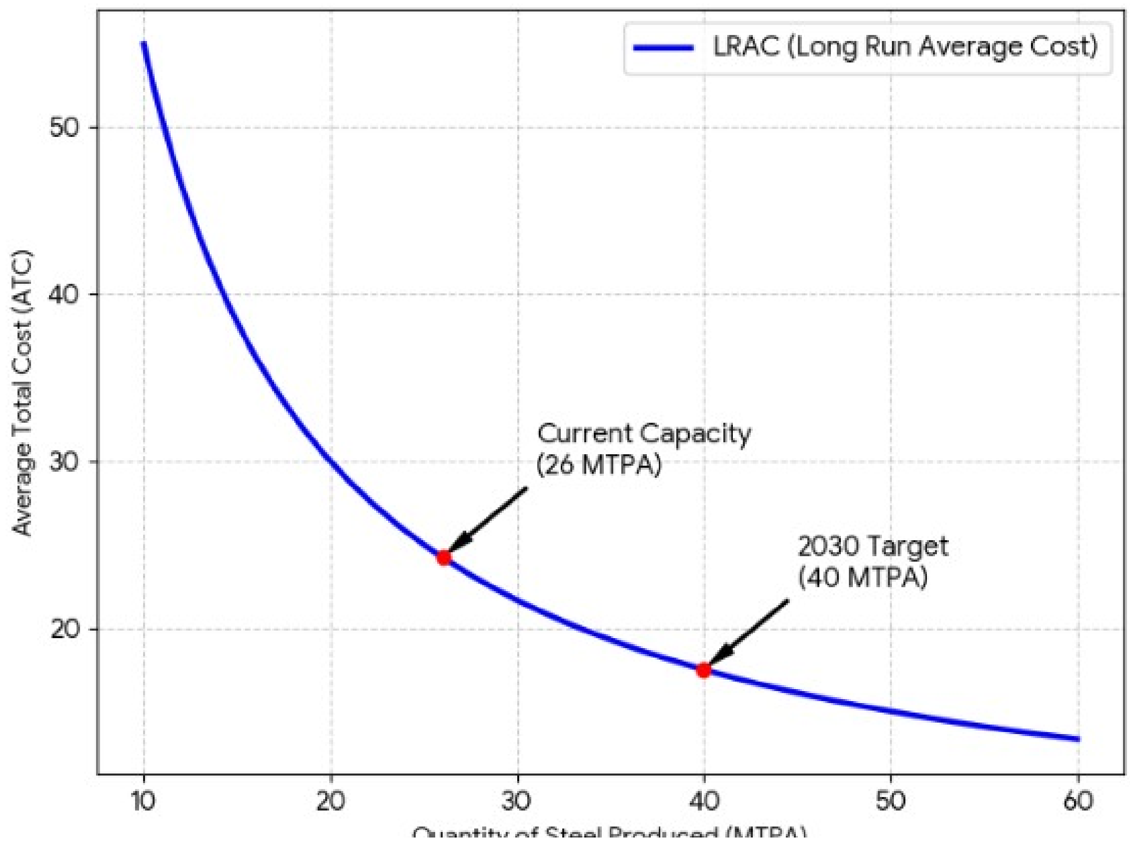
Tata Steel holds the position of second largest steel manufacturer in India while its market strength becomes evident through industrial analysis together with microeconomic evidence showing scale advantages and cost leadership and strategic product differentiation. The company was established in 1907 and its headquarters remain in Jamshedpur while it operates a 26 MTPA capacity across India at its major facilities located in Jamshedpur and Kalinganagar which produced 21.7 million tonnes of crude steel during FY25 while holding approximately 14.5% of the domestic market. The company demonstrates long-term supply expansion through its 2025–26 capital expenditure allocation of ₹15,000 crore which includes plans to increase its 40 MTPA production capacity through Kalinganagar and other facilities until 2030. The system operates through long-term supply expansion which decreases average total costs through economies of scale by distributing high fixed costs of blast furnaces and coke ovens and rolling mills to increased production levels which results in lower costs per unit.

Tata Steel achieves cost leadership through its vertical integration model which allows the company to control its resource management through captive iron ore mining combined with bulk procurement strategies that create input price stability. The firm maintains microeconomic efficiency by operating at minimum efficient scale which allows it to produce at 20–25% lower cost per tonne when compared to smaller companies while its energy optimization and gradual electric arc furnace (EAF) adoption lead to 10–15% reduction of variable costs. The company achieves 70% raw material self-sufficiency through backward integration which saves 10–14% in logistics costs, demonstrating how vertical control creates a downward supply curve shift which protects the company from input price changes.

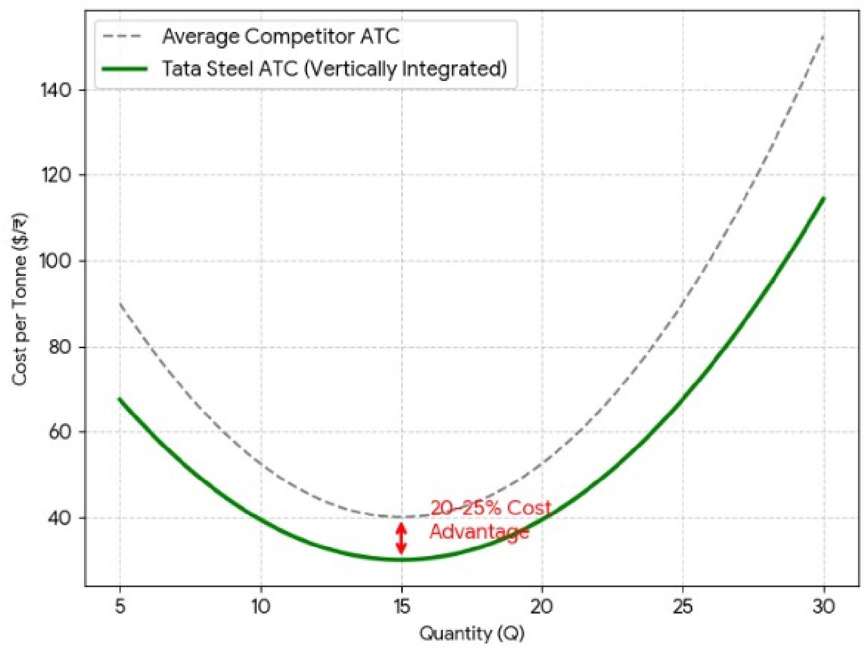
Tata Steel maintains market share through its long product pricing strategy which matches market prices because it operates in a special market where JSW Steel sells products. The market shows product differentiation through its niche segments which function at lower price elasticity because customers choose to buy products that provide consistent quality and technical specifications beyond price differences. The company achieved record Q3 FY26 deliveries of 6 MT which increased by 14% compared to last year while domestic sales accounted for approximately 72% of total sales, which supports its revenue growth through 8–9% internal demand increase that secures its national second position.

Profitability indicators confirm that microeconomic operations remain stable. The company achieved Q2 FY26 PAT of ₹3,102 crore which results from high capacity utilization together with operating leverage that drives revenue growth, while its improved return on equity shows better capital productivity from business expansion. The platform Aashiyana achieves GMV target of ₹7,000 crore through its downstream integration mechanism which establishes customer loyalty because it increases customer switching costs while enhancing market control over certain segments. Tata Steel uses its core drivers by combining its assets which include economies of scale and vertical integration and differentiated product mix and oligopolistic pricing discipline to achieve competitive advantage which will take the company to industry leadership by 2030.

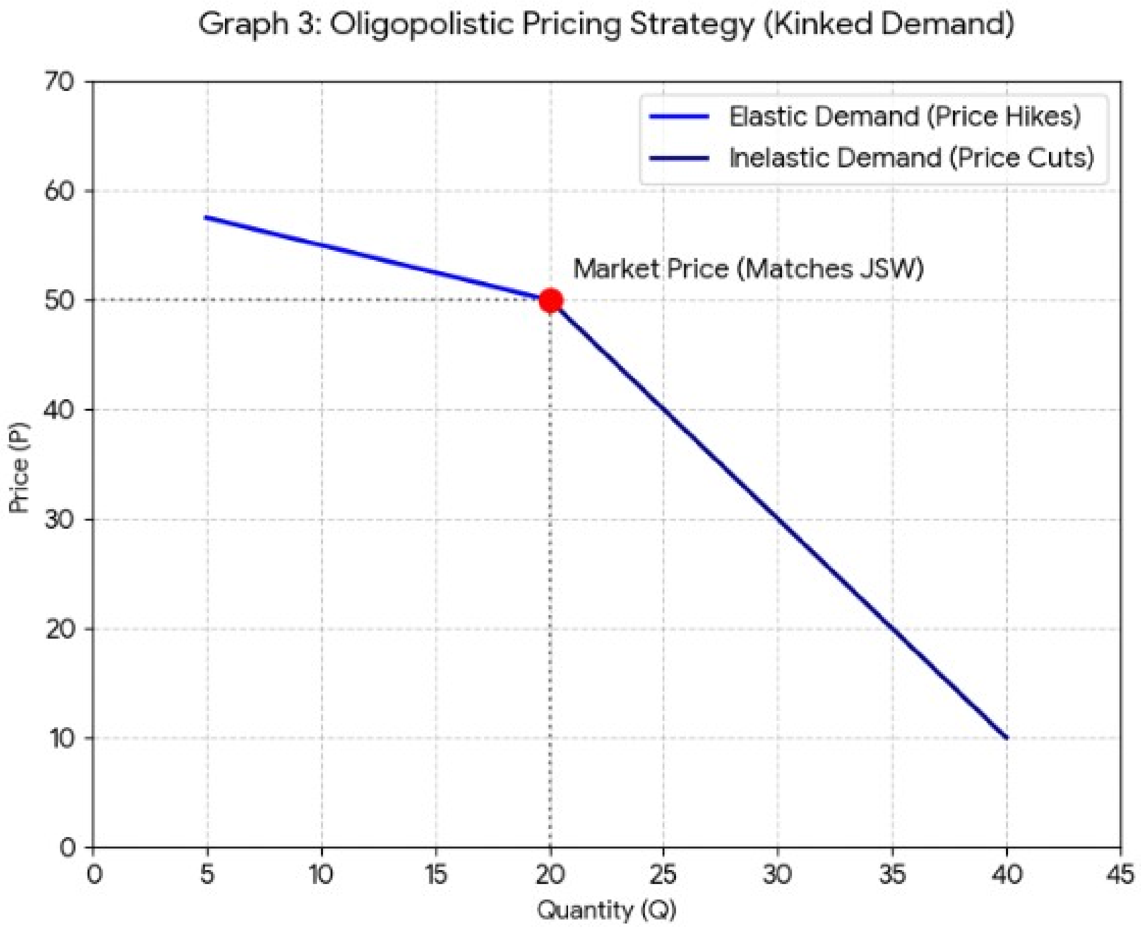
Graph 1: Economies of Scale (Tata Steel Expansion)



Graph 2: Cost Leadership via Vertical Integration



Avg
Total



Cost

Conclusion

India's iron and steel industry functions as a dynamic oligopoly system because it has a few powerful companies which control production and pricing and their long-term investments. The market structure creates high entry barriers which stem from the need for large capital investments and the requirement for direct access to iron ore and coking coal and the need for extensive logistics systems which operate at different scales. The market now sees companies choosing non-price competition methods which include building capacity and creating specialized steel products and developing green brand identities and establishing control over supply chains because price demand remains stable within short-term fluctuations between -0.2 and -0.5 for steel products which serve as an intermediate material vital for infrastructure development which accounts for 40 to 45 percent of the total use in cars and residential buildings and engineering projects. The income elasticity of 1.5 to 2.0

shows how steel consumption follows GDP growth patterns, meaning that Indian urbanization and capital expenditures should lead to increased steel demand as India approaches its target of becoming a $5 trillion economy.

The economic costs of the industry create conditions which maintain oligopoly stability because variable costs make up 60 to 70 percent of total company spending, while companies face challenges to control coal and iron ore price changes and their fixed costs require integrated plants to operate at high capacity levels between 85 and 90 percent for efficient overhead distribution. Tata Steel operates as the leading cost performer because it uses backward integration with captive mining and scale economies to achieve cost reductions between 20 and 25 percent while holding a 14.5 percent market share through its approach which includes expansion from 26 MTPA to 40 MTPA by 2030 to secure its future supply capability. The National Steel Policy 2017 targets of 8 to 9 percent demand growth until 2026 will be achieved through capacity expansion of 25 MTPA, which supports the implementation of Production Linked Incentive Scheme and specialty steel development according to industry forecasts. The situation faces three major challenges which include 75 to 80 percent utilization pressure, the risk of import surges, and export restrictions that arise from European Union Carbon Border Adjustment Mechanism (CBAM) regulations. The industry will shift to scrap-based electric arc furnaces and hydrogen pilot projects and Green Steel Mission emission reduction programs, which require a 10 to 15 percent reduction by 2030, so sustainability will become the key determining factor of future industry development. The industry operates under protection from established market structures and demand patterns and cost structures and policy framework development, yet it needs to develop new systems for carbon reduction and innovation to maintain its long-term market position.

Refrences

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